Eric Beaucé

Curriculum Vitae

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Academic Positions

09/2026 - present	**Research Scientist**	*Institut des Sciences de la Terre, Université Grenoble Alpes*
03/2026 - present	**Associate Research Scientist (on leave)**	*Lamont-Doherty Earth Observatory, Columbia University*
2022 - 2026	**Postdoctoral Brinson Fellow**	*Lamont-Doherty Earth Observatory, Columbia University*
09/2021 - 01/2022	**Postdoctoral Researcher**	*Massachusetts Institute of Technology*
2016 - 2021	**Research/Teaching assistant**	*Massachusetts Institute of Technology*

Ph.D. Thesis: Analyzing the Collective Behavior of Earthquakes to Understand Fault Mechanisms Better. Available at `https://tinyurl.com/EBPhDThesisManuscript`.
Supervised by Robert van der Hilst and Michel Campillo.

Education

2021	**Ph.D., Geophysics**	*Massachusetts Institute of Technology*
2016	**Master of Science, Physics**	*École Normale Supérieure de Lyon*
2014	**Bachelor of Science, Physics**	*École Normale Supérieure de Lyon*

Teaching Experience

2023 **Sonic and Visual Representation of Data** — *Columbia University*
Role: Teaching assistant. *Level*: Graduate.
Summary: Introduction to data sonification and visualization in Python.

2022 **Introduction to Statistical Seismology** — *Columbia University*
Role: Guest lecturer. *Level*: Graduate.

2021 **Introduction to Machine Learning in Earthquake Seismology** — *University of Colorado*
Role: Guest lecturer (remote). *Level*: Undergraduate.

2019 **Essentials of Geophysics** — *Massachusetts Institute of Technology*
Role: Teaching assistant. *Level*: Graduate.
Summary: Introduction to seismology, gravity, planetology, magnetism, and geodynamics.

2018 **Physical Principles of Remote Sensing** — *Massachusetts Institute of Technology*
Role: Teaching assistant. *Level*: Undergraduate.
Summary: Introduction to wave physics, Maxwell's equations, and their application to radar methods.

Field Experience

2022, 2023	**OBS deployment at the Axial Seamount**	*Pacific Coast, USA*
07/2018	**Preliminary passive seismic experiment (FaultProbe project)**	*San Jacinto, California, USA*
01/2018	**Groundwater flow imaging**	*Roseau Valley, Saint Lucia*
2016 - 2020	**Diverse subsurface exploration geophysical methods**	*New England, USA*

Grants and Fellowships

2026 - 2029 *"TideQuakes: Investigating the Preparation Phase of Large Earthquakes by Monitoring Tidally-Induced Earthquakes with Machine-Learning-Enhanced Catalogs"*
Choose France for Science (Agence Nationale de la Recherche), 215,200€
Eric Beaucé, (PI, ISTerre, UGA)

2026 - 2027 *"Investigating the Preparation Phase and Critical State Preceding Large Earthquakes Using the Tides as a Natural Probe"*
US Geological Survey, $88,790
Eric Beaucé (PI, LDEO), Felix Waldhauser (co-PI, LDEO)

2025 - 2027 *"Laboratory-driven, machine-learned constitutive models for fracture network evolution in the deep crust, with applications to extraction of thermal and chemical resources"*
Department of Energy, $76,862
Benjamin Holtzman (PI, MIT), **Eric Beaucé** (PI, LDEO), Folarin Kolawole (co-PI, LDEO)

2025 - 2027 *"Collaborative research: Investigating seimogenesis and faults in Eastern North America using a deep magnitude earthquake catalog"*
National Science Foundation, $319,877
Felix Waldhauser (PI, LDEO), **Eric Beaucé** (co-PI, LDEO)

2024 - 2027 *"Process-based experimental and machine learning approaches for controlling fracture network generation in the brittle-ductile crust"*
Department of Energy, $499,999
Seth Saltiel (PI, Cornell University), Benjamin Holtzman (PI, LDEO), **Eric Beaucé** (senior pers.)

2022 - 2025 *"Anticipating earthquakes initiative"*
Brinson Foundation postdoctoral fellowship
Eric Beaucé (fellow)

Outreach Activity

Seismic Sound Lab

The Seismic Sound Lab (`https://seismicsoundlab.github.io/`) engages audiences of all ages and backgrounds with the physics of earthquakes and seismic wave propagation by using sonified data as a communication tool. The Seismic Sound Lab plays a central role in Lamont-Doherty Earth Observatory's annual Open House and frequently presents its work to visitors on campus.

Conference and Seminar Organization

2026	Lamont Symposium on Artificial Intelligence-Based Science.
2025	AGU : Primary organizer of "*Collective Behaviors in Seismology: Models and Observations for Complexity in Seismicity, Crustal Mechanics, and Faulting*".
2023 - 2025	Organizer of weekly divisional seminars at Lamont.
2025	SSA : Co-organizer of workshop "*Building a High-Resolution Earthquake Catalog from Raw Waveforms: A Step-by-Step Guide*".
2025	Lamont Symposium on Earth Hazards : Moderator of "*Challenges in Forecasting Earth Hazards*".
2024	AGU : Co-organizer of "*On the Seismicity of "Stable" Continental Regions: The April 2024 New Jersey Earthquake and Beyond*".
2023	SSA : Primary organizer of "*Deciphering Earthquake Clustering for the Better Understanding of Crustal Deformation Mechanisms*".
2022	AGU : Co-organizer of "*Microseismicity and Fault Slip: Observations, Modeling, and Experiments*".

Invited Conference Talks and Seminars

- Brown University, Geophysics Seminar (USA, March 2026).
- 3rd Workshop on Earthquake Physics and Applications of AI to Tectonic Faulting (Italy, September 2025).
- UC Santa Cruz, Institute of Geophysics and Planetary Physics Seminar (USA, May 2025).
- US Geological Survey, Moffett Field, Earthquake Science Center Seminar (USA, May 2025).
- ERC TECTONIC / FEAR Seminars on Earthquake Physics (Online, November 2024).
- Massachusetts Institute of Technology, Geophysics Seminar (USA, November 2023).
- Ecole Normale Supérieure, Laboratoire de Géologie (France, October 2023).
- Los Alamos National Laboratory, Frontiers in Geoscience (USA, June 2023).

Submitted Articles

- **Eric Beaucé**, Piero Poli, Felix Waldhauser, Benjamin Holtzman and Chris Scholz. Shift and increase in tidal modulation of seismicity in the Ridgecrest fault zone before and after the 2019 M7.1 earthquake. *In review at Bulletin of Seismological Society of America*
- Won-Young Kim, **Eric Beaucé**, Felix Waldhauser and Folarin Kolawole. The Mw4.8 Tewksbury, New Jersey earthquake of 5 April 2024 likely triggered by transient stresses from rapidly refilled nearby reservoir. *In review at Geophysical Research Letters*

Articles in Preparation

- **Eric Beaucé** and Folarin Kolawole. Heterogeneous seismicity illuminates incipient rifting in Botswana.
- Anna Barth, **Eric Beaucé**, Benjamin Holtzman and Tushar Mittal. Seismicity across the supercritical-to-steam transition: Comparative patterns above two distinct heat sources in the Hengill Geothermal Field, Iceland.
- Farzaneh Mohammadi, **Eric Beaucé**, Romain Jolivet and Kristel Chanard. Seasonal modulation of seismicity in an intraplate setting, the case of southeastern Australia

Peer-reviewed Articles

2026

- Folarin Kolawole, **Eric Beaucé**, Zach Foster-Baril, Meritxell Colet, Leonardo Seeber, Jacob Tielke, Abhishek Prakash, Won-Young Kim, Rasheed Ajala, Christine McCarthy, Felix Waldhauser. The 2024 Mw4.8 New Jersey Intraplate Earthquake: Preferential Rupture of an Immature Fault in Frictionally Unstable Basement Rocks. *Journal of Geophysical Research: Solid Earth.* DOI: `https://doi.org/10.1029/2025JB033723`
- Rodrigo Flores Allende, Léonard Seydoux, **Eric Beaucé**, Luis Fabian Bonilla, Philippe Gueguen, Claudio Satriano. Fine-Scale Segmentation and Spatiotemporal Variability of the 2010 Mw 8.8 Maule Aftershock Sequence Revealed by a Deep-Learning-Based Earthquake Catalog. *Journal of Geophysical Research: Solid Earth.* DOI: `https://doi.org/10.1029/2026JB034262`

2025

- **Eric Beaucé**. Measuring and modelling the occupation probability to characterize the temporal statistics of seismic sequences. *Geophysical Journal International.* DOI: `https://doi.org/10.1093/gji/ggaf433`
- **Eric Beaucé**, Felix Waldhauser, David Schaff, Won-Young Kim, Folarin Kolawole. The 5 April 2024 M_w4.8 Tewksbury, New Jersey aftershock sequence resolved with machine-learning-enhanced detection methods. *Geophysical Research Letters.* DOI: `https://doi.org/10.1029/2024GL113598`

2024

- Tanner Acquisto, Anne Bécel, Juan Pablo Canales, **Eric Beaucé**. Structural controls on megathrust slip behavior inferred from a 3D, crustal-scale, P-wave velocity model of the Alaska Peninsula subduction zone. *Journal of Geophysical Research: Solid Earth*. DOI: `https://doi.org/10.1029/2024JB029632`
- Jens-Erik Lundstern, **Eric Beaucé** and Orlando J. Teran. The Importance of Nodal Plane Orientation Diversity for Earthquake Focal Mechanism Stress Inversions. *Geological Society of London*. DOI: `https://doi.org/10.1144/SP546-2023-63`.

2023

- **Eric Beaucé**, Piero Poli, Felix Waldhauser, Benjamin Holtzman, and Chris Scholz. Enhanced tidal sensitivity of seismicity before the 2019 M7.1 Ridgecrest, CA earthquake. *Geophysical Research Letters*. DOI: `https://doi.org/10.1029/2023GL104375`.
- **Eric Beaucé**, William B. Frank, Léonard Seydoux, Piero Poli, Nathan Groebner, Robert D. van der Hilst and Michel Campillo. BackProjection and Matched-Filtering (BPMF): An Automated Earthquake Detection and Location Workflow. *Seismological Research Letters: Electronic Seismologist*. DOI: `https://doi.org/10.1785/0220230230`.

2022

- **Eric Beaucé**, Robert D. van der Hilst, Michel Campillo. Microseismic Constraints on the Mechanical State of the North Anatolian Fault Thirteen Years after the 1999 M7.4 Izmit Earthquake. *Journal of Geophysical Research: Solid Earth*. DOI: `https://doi.org/10.1029/2022JB024416`.
- **Eric Beaucé**, Robert D. van der Hilst, Michel Campillo. An Iterative Linear Method with Variable Shear Stress Magnitudes for Estimating the Stress Tensor from Earthquake Focal Mechanism Data: Method and Examples. *Bulletin of the Seismological Society of America*. DOI: `https://doi.org/10.1785/0120210319`.
- René Steinmann, Léonard Seydoux, **Eric Beaucé**, Michel Campillo. Hierarchical Exploration of Continuous Seismograms with Unsupervised Learning. *Journal of Geophysical Research: Solid Earth*. DOI: `https://doi.org/10.1029/2021JB022455`.

2021

- Hugo Sanchéz-Reyes, David Essing, **Eric Beaucé**, Piero Poli. The Imbricated Foreshock and Aftershock Activities of the Balsorano (Italy) Mw 4.4 Normal Fault Earthquake and Implications for Earthquake Initiation. *Seismological Research Letters*. DOI: `https://doi.org/10.1785/0220200253`.

2019

- **Eric Beaucé**, William B. Frank, Anne Paul, Michel Campillo and Robert D. van der Hilst. Systematic Detection of Clustered Seismicity beneath the Southwestern Alps. *Journal of Geophysical Research: Solid Earth*. DOI: `http://dx.doi.org/10.1029/2019JB018110`.
- Florent Brenguier, Pierre Boué, Yehuda Ben-Zion, F. Vernon, C.W. Johnson, A. Mordret, O. Coutant, P-E. Share, **Eric Beaucé**, D. Hollis, T. Lecocq. Train Traffic as a Powerful Noise Source for Monitoring Active Faults with Seismic Interferometry. *Geophysical Research Letters*. DOI: `http://dx.doi.org/10.1029/2019GL083438`.

2017

- **Eric Beaucé**, William B. Frank and Alexey Romanenko. Fast Matched Filter (FMF): An Efficient Seismic Matched-Filter Search for Both CPU and GPU Architectures. *Seismological Research Letter*. DOI: `https://doi.org/10.1785/0220170181`.